

Quantitative Aptitude

Number System

STUDY NOTES

Square and Square Roots



Square and Square Roots

Square

The **square** of a number is the result of multiplying that number by itself. In mathematical terms, the square of a number X is written as x^2 or $X \times X$. The operation of squaring is an important concept in arithmetic, algebra, and geometry and is frequently used in various mathematical calculations.

Formula:

If X is a number, then its square is:

$$X^2 = X \times X$$

Examples:

- The square of 3 is $3^2 = 3 \times 3 = 9$
- The square of 5 is $5^2 = 5 \times 5 = 25$.

Properties of Square Numbers:

- **Positive Result:** The square of any real number (positive or negative) is always positive or zero.
e.g., $5^2 = 5 \times 5 = 25$ or $(-5)^2 = (-5) \times (-5) = 25$, the result will always be positive.
- **Perfect Squares:** Numbers like 1, 4, 9, 16, etc., are called perfect squares because they are the squares of whole numbers.
- **Exponential Representation:** The square of a number is represented by raising it to the power of 2, i.e., x^2 .
- All square numbers end with 0, 1, 4, 5, 6 or 9 at the unit's place. Square numbers can only have an even number of zeros at the end.
- Square root is the inverse operation of square. There are two integral square roots of a perfect square number.
- All these numbers end with 0, 1, 4, 5, 6 or 9 at the unit's place. None of these end with 2, 3, 7 or 8 at the unit's place.

Example 1: Which of the following numbers is not a square number?

- a) 225 b) 625 c) 441 d) 741

Solution:

$$\Rightarrow 225 = 15^2$$

$$\Rightarrow 625 = 25^2$$

$$\Rightarrow 441 = 21^2$$

$\Rightarrow 741$ does not equal n^2 for any integer n .

Hence, The answer of the option d.

Example 2: Which of the following is not a perfect square?

- a) 576 b) 841 c) 778 d) 529

Solution:

Option a: $\sqrt{576} = 24$

Option b: $\sqrt{841} = 29$

Option c: $\sqrt{778} = 27.89$ It is not a perfect square

Option d: $\sqrt{529} = 23$

$\therefore$ The correct answer is 741.

Example 3: The following numbers are not perfect squares. Give a reason.

i. 1047

ii. 23543

iii. 7628

iv. 332222

v. 65000

vi. 87222

vii. 333000

viii. 404040

Solution:

We know that natural numbers ending in the digits 0, 2, 3, 7 and 8 are not perfect squares.

- i. 1047 Ends with 7
- ii. 23543 Ends with 3
- iii. 7628 Ends with 8
- iv. 332222 Ends with 2
- v. 65000 Ends with 0
- vi. 87222 Ends with 2
- vii. 333000 Ends with 0
- viii. 404040 Ends with 0

Square of a number with unit's place 5:

The formula for squaring a number ending in 5, say $n = 10x + 5$, is:

$N^2 = (x)(x + 1)$ and then add 25

where N can have one or more than one digits.

Steps:

1. **Remove the 5** from the number, leaving the remaining digits (denoted x).
2. **Multiply x by $x + 1$.**
3. **Place 25 at the end** of the result from step 2.

Examples:

1. **Square of 25:**
 - **Step 1:** Remove the 5, leaving $x = 2$.
 - **Step 2:** $x \times (x+1) = 2 \times 3 = 6$
 - **Step 3:** Append 25 $\rightarrow$ The square of 25 is 625.

2. Square of 35:

- **Step 1:** Remove the 5, leaving $x = 3$.
- **Step 2:** $x \times (x+1) = 3 \times 4 = 12$.
- **Step 3:** Append 25 $\rightarrow$ The square of 35 is 1225.

3. Square of 105:

- **Step 1:** Remove the 5, leaving $x = 10$.
- **Step 2:** $x \times (x+1) = 10 \times 11 = 110$.
- **Step 3:** Append 25 $\rightarrow$ The square of 105 is 11025

Example 4: Finding the Square of a Number 35.**Solution:**

The last two digits are the square of 5. That is 25

For first two digit

$$\text{Add} = 3 + 1 = 4$$

$$\text{Multiply } 3 \times 4 = 12$$

We get the square of 35 = 1225

$$35^2 = 1225$$

$\therefore$ The correct answer is 1225.

Example 5: $115^2 = ?$ **Solution:**

The last two digits are the square of 5. That is 25

For the first three-digit

$$\text{Add} = 11 + 1 = 12$$

$$\text{Multiply } 11 \times 12 = 132$$

$\therefore$ The correct answer is 13225.

Example 6: The square of 53 is

Solution:

The square of 53 can be found by multiplying 53 by itself.

$$\Rightarrow 53 \times 53 = 2809$$

$$53^2 = 2809$$

$$25 + 3 = 3$$

$$3^2 = 9$$

$$2809$$

∴ The correct answer is 2809.

Random Interesting Patterns Followed by Square Numbers

Patterns in numbers like 1, 11, 111, ... :

$$1^2 = 1$$

$$11^2 = 121$$

$$111^2 = 12321$$

$$1111^2 = 1234321$$

$$11111^2 = 123454321$$

$$11111111^2 = 123456787654321$$

Patterns in numbers like 6, 67, 667, ... :

$$7^2 = 49$$

$$67^2 = 4489$$

$$667^2 = 444889$$

$$6667^2 = 44448889$$

$$66667^2 = 4444488889$$

$$666667^2 = 444444888889$$

Cube:

The **cube of a number** is the result of multiplying that number by itself twice, or, in other words, raising the number to the power of three. For a given number X , the cube is written as x^3 , meaning $X \times X \times X$.

Formula:

If x is any number, then: Cube of $x = x^3 = X \times X \times X$

Examples:

- The cube of 2 is $2^3 = 2 \times 2 \times 2 = 8$
- The cube of 5 is $5^3 = 5 \times 5 \times 5 = 125$

Properties of Cubed Numbers:

1. **Positive or Negative:** The cube of a positive number is positive (e.g., $3^3 = 27$, and the cube of a negative number is negative (e.g., $(-3)^3 = -27$).
2. **Perfect Cubes:** Numbers like 1, 8, 27, 64, etc., are called **perfect cubes** because they are the cubes of whole numbers.

Squares and Cubes table

The **list of squares and cubes up to 40** is given in the table below.

Squares (n^2)		Cubes (n^3)	
$1^2 = 1$	$21^2 = 441$	$1^3 = 1$	$21^3 = 9261$
$2^2 = 4$	$22^2 = 484$	$2^3 = 8$	$22^3 = 10648$
$3^2 = 9$	$23^2 = 529$	$3^3 = 27$	$23^3 = 12167$

$4^2 = 16$	$24^2 = 576$	$4^3 = 64$	$24^3 = 13824$
$5^2 = 25$	$25^2 = 625$	$5^3 = 125$	$25^3 = 15625$
$6^2 = 36$	$26^2 = 676$	$6^3 = 216$	$26^3 = 17576$
$7^2 = 49$	$27^2 = 729$	$7^3 = 343$	$27^3 = 19683$
$8^2 = 64$	$28^2 = 784$	$8^3 = 512$	$28^3 = 21952$
$9^2 = 81$	$29^2 = 841$	$9^3 = 729$	$29^3 = 24389$
$10^2 = 100$	$30^2 = 900$	$10^3 = 1000$	$30^3 = 27000$
$11^2 = 121$	$31^2 = 961$	$11^3 = 1331$	$31^3 = 29791$
$12^2 = 144$	$32^2 = 1024$	$12^3 = 1728$	$32^3 = 32768$
$13^2 = 169$	$33^2 = 1089$	$13^3 = 2197$	$33^3 = 35937$
$14^2 = 196$	$34^2 = 1156$	$14^3 = 2744$	$34^3 = 39304$
$15^2 = 225$	$35^2 = 1225$	$15^3 = 3375$	$35^3 = 42875$

$16^2 = 256$	$36^2 = 1296$	$16^3 = 4096$	$36^3 = 46656$
$17^2 = 289$	$37^2 = 1369$	$17^3 = 4913$	$37^3 = 50653$
$18^2 = 324$	$38^2 = 1444$	$18^3 = 5832$	$38^3 = 54872$
$19^2 = 361$	$39^2 = 1521$	$19^3 = 6859$	$39^3 = 59319$
$20^2 = 400$	$40^2 = 1600$	$20^3 = 8000$	$40^3 = 64000$

Square Root:

The **square root** of a number is a value that, when multiplied by itself, gives the original number. In mathematical terms, if X is the square root of Y, then $X^2 = Y$. The square root of Y is often denoted by the symbol $\sqrt{Y}$.

Formula:

For a number Y, the square root is:

$$X = \sqrt{Y}$$

where $X \times X = y$

Examples:

- The square root of 16 is 4, because $4 \times 4 = 16$
- The square root of 25 is 5, because $5 \times 5 = 25$

Properties of Square Roots:

1. **Non-negative Result:** The principal square root (positive square root) of a positive number is always non-negative. For example, $\sqrt{9} = 3$ and not -3 , even though $(-3)^2 = 9$
2. **Perfect Squares:** Numbers like 1, 4, 9, 16, etc., are called **perfect squares** because they have whole number square roots.
3. **Irrational Square Roots:** For numbers that are not perfect squares (e.g., 2, 3, 5), the square roots are irrational, meaning they have non-repeating, non-terminating decimal values (e.g., $\sqrt{2} \approx 1.414$).

We have, $1^2 = 1$, therefore square root of 1 is 1

$2^2 = 4$, therefore square root of 4 is 2

$3^2 = 9$, therefore square root of 9 is 3

Statement	Inference	Statement	Inference
$6^2 = 36$	$\sqrt{36} = 6$	$1^2 = 1$	$\sqrt{1} = 1$
$7^2 = 49$	$\sqrt{49} = 7$	$2^2 = 4$	$\sqrt{4} = 2$
$8^2 = 64$	$\sqrt{64} = 8$	$3^2 = 9$	$\sqrt{9} = 3$
$9^2 = 81$	$\sqrt{81} = 9$	$4^2 = 16$	$\sqrt{16} = 4$
$10^2 = 100$	$\sqrt{100} = 10$	$5^2 = 25$	$\sqrt{25} = 5$

There are several methods for finding the square root of a number, each useful for different contexts. Here are some common methods:

Prime Factorization Method

- ❖ **Description:** This method is useful for finding the square roots of perfect squares by breaking down the number into its prime factors.
- ❖ **Steps:**
 - Find the prime factorization of the number.
 - Pair the prime factors in groups of two.

➤ Take one number from each pair and multiply them.

❖ **Example:** To find the square root of 144:

➤ Prime factors of 144 are $2 \times 2 \times 2 \times 2 \times 3 \times 3$

➤ Pair the factors: $(2 \times 2)(2 \times 2)$ and (3×3)

➤ Square root of 144 = $2 \times 2 \times 3 = 12$

Example 7: Find the square root of 6400.

Solution:

2	6 4 0 0
2	3 2 0 0
2	1 6 0 0
2	8 0 0
2	4 0 0
2	2 0 0
2	1 0 0
2	5 0
5	2 5
	5

Write $6400 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 5$

By pairing the prime factors, we get

Therefore, $6400 = 2 \times 2 \times 2 \times 2 \times 5 = 80$

∴ **The correct answer is 80.**

Example 8: Is 2352 a perfect square? If not, find the smallest multiple of 2352 which is a perfect square. Find the square root of the new number.

2	2 3 5 2
2	1 1 7 6
2	5 8 8
2	2 9 4
3	1 4 7
7	4 9
	7

We have $2352 = 2 \times 2 \times 2 \times 2 \times 3 \times 7 \times 7$

As the prime factor 3 has no pair, 2352 is not a perfect square.

If 3 gets a pair then the number will become a perfect square. So, we multiply 2352 by 3 to get,

$$2352 \times 3 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 7 \times 7$$

Now each prime factor is in a pair. Therefore, $2352 \times 3 = 7056$ is a perfect square.

Thus the required smallest multiple of 2352 is 7056 which is a perfect square.

$$\text{And, } 7056 = 2 \times 2 \times 3 \times 7 = 84$$

$\therefore$ The correct answer is 84.

Example 9: Find the smallest number by which 9408 must be divided so that the quotient is a perfect square. Find the square root of the quotient.

Solution:

$$\text{We have, } 9408 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 7 \times 7$$

If we divide 9408 by the factor 3, then

$$9408 \div 3 = 3136 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 7 \times 7 \text{ which is a perfect square.}$$

Therefore, the required smallest number is 3.

$$\text{And, } 3136 = 2 \times 2 \times 2 \times 7 = 56.$$

$\therefore$ The correct answer is 56.

Example 10: Square root of 1296 is:

Solution:

Given number is 1296,

Prime factors of 1296:

$$1296 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3$$

$$1296 = 2^2 \times 2^2 \times 3^2 \times 3^2$$

$$\text{Square root of } 1296 = 2 \times 2 \times 3 \times 3 = 36.$$

∴ The correct answer is 36.

Example 11: Square root of 260100 is:

Solution:

Given number is 260100,

Prime factors of 260100:

$$260100 = 2 \times 2 \times 3 \times 3 \times 5 \times 5 \times 17 \times 17$$

$$260100 = 2^2 \times 3^2 \times 5^2 \times 17^2$$

$$\text{Square root of } 260100 = 2 \times 3 \times 5 \times 17 = 510.$$

∴ The correct answer is 510.

Long Division Method

Steps involved in finding the square root of 484 by Long division method:

Step 1: Place a bar over every pair of numbers starting from the digit at units place. If the number of digits in it is odd, then the left-most single-digit too will have a bar.

Step 2: Take the largest number as divisor whose square is less than or equal to the number on the extreme left. Divide and write quotient.

$$\begin{array}{r} 2 \\ 2 \overline{) 484} \\ \underline{-4} \\ 0 \end{array}$$

Step 3: Bring down the number which is under the next bar to the right side of the remainder.

$$\begin{array}{r} 2 \\ 2 \overline{) 484} \\ \underline{-4} \\ 084 \end{array}$$

Step 4: Double the value of the quotient and enter it with a blank on the right side.

$$\begin{array}{r} 2 \\ 2 \overline{) 484} \\ \underline{-4} \\ 4-084 \end{array}$$

Step 5: Guess the largest possible digit to fill the blank which will also become the new digit in the quotient, such that when the new divisor is multiplied to the new quotient the product is less than or equal to the dividend.

$$\begin{array}{r} 2 \quad 2 \\ 2 \overline{) 484} \\ \underline{-4} \\ 42-084 \\ \underline{-84} \\ 0 \end{array}$$

Square root of 484 = 22.

Steps to Find Square Root of 144 by Division Method:

Step 1: Group the digits

- First, pair the digits of the number starting from the decimal point, moving left and right. For 144, pair the digits as (1) (44).

Step 2: Find the largest number whose square is less than or equal to the first pair

- The first pair is "1". The largest number whose square is less than or equal to 1 is 1 (because $1^2 = 1$). Write 1 as the divisor and the quotient.

Step 3: Perform the division

- Divide 1 by 1. This gives a quotient of 1, and the remainder is 0. Now, bring down the next pair of digits, "44", next to the remainder (which is 0).

Step 4: Double the quotient

- Double the quotient (which is 1) to get 2. Write this doubled value as a part of the new divisor, leaving a blank space next to it (let's call it 2_).

Step 5: Find the next digit in the quotient

- Now, you have 44 to divide. To find the next digit, choose a number that when placed in the blank space next to "2" forms a new divisor and can divide into 44.
 - Try 2. Place 2 next to "2" to make 22.
 - Divide 44 by 22, which gives 2. Now, the quotient becomes 12, and the divisor is 22.

Step 6: Perform the division

- Divide 44 by 22 to get a quotient of 2, and the remainder is 0.

Step 7: Stop when the remainder is zero

- Since there are no more digits to bring down, and the remainder is 0, the process is complete.

So, the square root of 144 is 12.

$$\begin{array}{r}
 12 \\
 1 \overline{) 144} \\
 + 1 \overline{) -144} \\
 \hline
 22 \overline{) 044} \\
 + 2 \overline{) -44} \\
 \hline
 0
 \end{array}$$

The square root of Decimals

Step 1. Consider the number 24.01 whose square root is to be calculated. We put the bar on the integral part of the number starting from the unit's digit. This bar is placed in pairs. Similarly for decimal part, we put the bar on the pair of numbers starting from the digit which is immediately placed after the decimal. In this case, it will be $\overline{24} . \overline{01}$.

Step 2. The divisor will be the maximum number whose square is less than or equal to the dividend that is the integral part of the number. In this case the divisor will be 4 and the dividend is 24. Divide and get the remainder.

$$\begin{array}{r}
 4 \\
 4 \overline{) 24 \overline{01}} \\
 - 16 \\
 \hline
 8
 \end{array}$$

Step 3. The remainder in this case is 8. Write the number which is in the next bar to the right side of the remainder. We get 801.

Step 4. Double the value of the divisor and enter it with a blank on its right. Since 01 is the decimal part of the number, put a decimal point in the quotient.

$$\begin{array}{r}
 4 \overline{) 24 \overline{01}} \\
 - 16 \\
 \hline
 8- \overline{) 801}
 \end{array}$$

Step 5. Now we have to select the largest digit for the unit place of the divisor (8_) such that the new number when multiplied by the new digit at unit's place is equal to or less than the dividend (801). As we can see that $89 \times 9 = 801$, the new digit of the divisor is 9 and thus the divisor is 89. Divide the number and get the remainder.

$$\begin{array}{r}
 4 \\
 4 \overline{) 2401} \\
 \underline{-16} \\
 801
 \end{array}$$

Step 6. Since the remainder is 0 and no bar is left,

$$\begin{array}{r}
 4.9 \\
 4 \overline{) 2401} \\
 \underline{-16} \\
 801 \\
 89 \overline{) 801} \\
 \underline{-80} \\
 1
 \end{array}$$

Therefore we can say that $\sqrt{24.01} = 4.9$

Example 12: Find the least number that must be subtracted from 5607 so as to get a perfect square. Also find the square root of the perfect square.

Solution:

Let us try to find $\sqrt{5476}$ by long division method. We get the remainder 131. It shows that 74^2 is less than 5607 by 131.

This means if we subtract the remainder from the number, we get a perfect square.

Therefore, the required perfect square is $5607 - 131 = 5476$.

And, $\sqrt{5476} = 74$.

∴ The correct answer is 74.

$$\begin{array}{r}
 74 \\
 7 \overline{) 5607} \\
 \underline{-49} \\
 144 \overline{) 707} \\
 \underline{-576} \\
 131
 \end{array}$$

Example 13: Find the greatest 4-digit number which is a perfect square.

Solution:

Greatest number of 4-digits = 9999. We find $\sqrt{9999}$ by the long division method.

The remainder is 198. This shows 99^2 is less than 9999 by 198.

99	
9	99 99 - 81
189	1 8 9 9 - 1 7 0 1
	1 9 8

This means if we subtract the remainder from the number, we get a perfect square.

Therefore, the required perfect square is $9999 - 198 = 9801$.

$$\sqrt{9801} = 99$$

$\therefore$ The correct answer is 99.

Example 14: Find the least number that must be added to 1300 so as to get a perfect square. Also find the square root of the perfect square.

Solution:

We find 1300 by long division method. The remainder is 4.

This shows that $36^2 < 1300$.

Next perfect square number is $37^2 = 1369$.

Hence, the number to be added is $37^2 - 1300 = 1369 - 1300 = 69$.

$$\begin{array}{r}
 36 \\
 3 \overline{) 1300} \\
 \underline{-9} \\
 66 \\
 \underline{-66} \\
 00 \\
 \underline{-00} \\
 00 \\
 \underline{-00} \\
 00
 \end{array}$$

Example 15: Find $\sqrt{12.25}$

Step 1: Pair the digits starting from the decimal point

- Just like for whole numbers, start pairing the digits from the decimal point, moving left for the whole part and right for the decimal part.
- For 12.25, we group the digits as (12) and (25).

Step 2: Find the largest number whose square is less than or equal to the first pair

- The first pair is "12". The largest number whose square is less than or equal to 12 is 3 (because $3^2 = 9$). Write 3 as the quotient and the divisor.

Step 3: Perform the division

- Divide 12 by 3. This gives a quotient of 3, and the remainder is $12 - 9 = 3$. Now, bring down the next pair of digits (which is "25").

Step 4: Double the quotient

- Double the quotient (which is 3) to get 6. Write this doubled value as part of the new divisor, leaving a blank space next to it (let's call it 6_).

Step 5: Add a decimal point to the quotient

- Add a decimal point to the quotient at this stage. The next number in the quotient will appear after the decimal point.

Step 6: Bring down the next pair of digits

- Bring down the next pair of digits, which is "25", making the current number 325.

Step 7: Find the next digit in the quotient

- Now, choose a number that, when placed next to "6" in the divisor, forms a new divisor that can divide into 325.
 - Try 5. Place 5 next to 6 to make the new divisor 65.
 - Divide 325 by 65, which gives a quotient of 5.

Step 8: Perform the division

- Divide 325 by 65, which gives the quotient 5, and the remainder is 0.

Step 9: Stop when remainder is zero

- Since there are no more digits to bring down, and the remainder is 0, the process is complete.

So, the square root of 12.25 is 3.5.

	3.5	
3	12.25	
	9	↓
65	325	
	325	
	0	

Example 16: What is the value of $\sqrt{(0.000081)}$?

Solution:

Formula used:

$$\sqrt{(a \times 10^b)} = \sqrt{a} \times 10^{b/2}$$

Calculation:

0.000081 can be written as 81×10^{-6}

$$\text{So, } \sqrt{(0.000081)} = \sqrt{(81 \times 10^{-6})}$$

$$\Rightarrow \sqrt{81} \times \sqrt{10^{-6}}$$

$$\Rightarrow 9 \times 10^{-3}$$

$$\Rightarrow 9 \times 0.001$$

$$\Rightarrow 0.009$$

∴ The correct answer is 0.009.

Example 17: Find the square root of 6466849

Solution:

Calculation:

2	6 4 6 6 8 4 9	
+ 2	4	2 5 4 3
4 5	2 4 6	
+ 5	2 2 5	
5 0 4	2 1 6 8	
+ 4	2 0 1 6	
5 0 8 3	1 5 2 4 9	
+ 3	1 5 2 4 9	
5 0 8 6	X	

The square root of 6466849 = 2543.

∴ The correct answer is 2543.